MMCTL Manuscript Change Highlight

Change date: 2026-09-20

Baseline: bare_jrnl_original_baseline.tex (first committed draft)
Current: bare_jrnl_new_sample4.tex (revision with B1–B3 edits)
Red = added / changed lines **Green** = removed baseline lines (context)

Reviewer-comment status snapshot

| R1.1 | Clarify essential difference vs supervised contrastive / cross-modal triplet; continuous similarity contribution | **DONE** | Related Work + contribution bullets (i)–(iv); intro inductive-bias framing |

| R1.2 | Interpretable latent space: visualizations or cross-modal retrieval | **PARTIAL** | Existing PCA fig + new failure/retrieval fig + class R1 80.8% / identity 7.5%. Successes grid not placed in PDF (space). Optional: add successes fig if pages allow |

| R1.3 | Specify pos/neg selection rules; similarity thresholds / margins | **DONE** | Method: batch-all; order-only criterion; no hard threshold; margin-free adaptive gap |

| R1.4 | How annotations normalized/weighted; does weighting affect results? | **DONE** | Eqs. for norm/weight; class-weight ablation Fig. already in paper |

| R1.5 | Rationale for freezing encoder; compare end-to-end fine-tuning | **PARTIAL** | Freeze framed as fair fusion protocol. **OPEN:** freeze vs E2E experiment (skipped; no checkpoints / optional) |

| R1.6 | Stronger baselines: SupCon, cross-attention, Transformer fusion | **OPEN** | Not run. Need at least one stronger baseline for a complete response |

| R1.7 | Report std + statistical significance | **DONE** (AWIR) | Wilcoxon n=8, structure p=0.0078; proxy downstream p≤0.016. **OPEN:** NEON significance once NEON replicated |

| — | Overall “method too simple / limited contribution” | **PARTIAL** | Related Work + incremental bullets. Still need NEON repro + ideally R1.5/R1.6 for strength |

| R2.1 | State substantive increment vs prior work (Eqs 1–3, continuous triplet, reg, multimodal extension) | **DONE** | Related Work + explicit contribution bullets tied to eqs |

| R2.2 | Intro uses gradient conflict without experiments — revise motivation | **DONE** | Intro rewritten; gradient-conflict claim removed |

| R2.3 | Discuss vs soft/continuous similarity ML and adaptive-margin triplets | **DONE** | Related Work (Kim 2019, Ha adaptive-margin, etc.) |

| R2.4 | Add cross-dataset/cross-region transfer **or** missing-modality | **PARTIAL** | AWIR missing-modality probes done. **OPEN:** cross-region / NEON transfer; NEON missing-modality |

| R2.5 | Report #triplets, time/epoch, GPU memory (batch-all) | **PARTIAL** | Upper-bound triplets + M4 microbench. **OPEN:** full schedule wall-clock / true peak GPU memory on training GPU |

| R2.6 | Clarify k-fold vs “averaged over 5 runs” in Fig. 3 caption | **DONE** | Caption aligned with 5 NEON CV folds |

| R2.7 | Error analysis: when method works / fails | **PARTIAL** | AWIR retrieval failure analysis. **OPEN:** richer NEON failure cases; more typical downstream failure discussion |

| R2.8 | Update refs to formally published versions (e.g. [8]) | **PARTIAL** | `zhou2025multi` → IGARSS 2025; added Khosla/Ha. **OPEN:** audit remaining arXiv-only cites (`reiter2020deep` still arXiv) |

| AWIR structure table | **DONE — PASS** | Keep paper numbers; max drift 0.13 |

| AWIR downstream table | **PARTIAL** | Keep paper absolutes; proxy cls OK; DeepSet proxy FAIL |

| AWIR missing modality | **DONE** | In manuscript + `missing_modality.json` |

| AWIR significance | **DONE** | In manuscript + `significance.json` |

| AWIR compute cost | **PARTIAL** | Upper bound + microbench in manuscript |

| AWIR failure/retrieval | **DONE** | In manuscript + figures |

| **NEON structure table** | **OPEN / IN PROGRESS** | No pretrained NEON embeddings in repo; proxy training required |

| **NEON downstream table** | **OPEN / IN PROGRESS** | Same |

| NEON significance | **OPEN** | Depends on NEON repro |

| Stronger baseline (R1.6) | **OPEN** | Recommended before resubmit |

| Freeze vs E2E (R1.5) | **OPEN** | Recommended if compute allows |

| Cross-region transfer | **OPEN** | Alternative/addition to missing-mod for R2.4 |

| PDF ≤8 pages verified | **OPEN** | Estimate ~7.5–8; need local `pdflatex` compile |

| Intro rewrite | **DONE** |

| Related Work + contributions | **DONE** |

| Method clarifications (triplets/margin/weights/freeze) | **DONE** |

| Same-labels / different-bias claim | **DONE** |

| Reproducibility paragraph | **DONE** |

| Significance paragraph | **DONE** |

| Missing-modality paragraph | **DONE** |

| Compute-cost paragraph | **DONE** |

| Failure/retrieval subsection + fig | **DONE** |

| Fig. ablation caption fix | **DONE** |

| LaTeX `\err{0.88}` fix | **DONE** |

| Bibliography updates | **PARTIAL** |

| Response-to-reviewers letter | **OPEN** | Draft after remaining experiments |

| Final page-budget trim | **OPEN** | After NEON + optional experiments |

Line-level TeX diff (red = new/changed)

Only changed hunks are shown. Unchanged context lines are gray.

Hunk baseline L83–85 → current L83–85 (replace)

```
– Supervised contrastive learning and cross-modal triplets typically define positives and negatives with binary
co-membership (same class or matched view) and a fixed similarity threshold or margin
\cite{khosla2020supervised,dutt2022shared}.
–
– MMCTL differs along four axes. (i)~A joint annotation vector with feature-wise normalization, task-length
rescaling, and fixed task weights (Eqs.~\eqref{eq:ann-norm}--\eqref{eq:ann-concat}) defines a continuous
multi-source similarity space
+ Binary supervised contrastive and cross-modal triplets define positives by class or view co-membership with a
fixed margin \cite{khosla2020supervised,dutt2022shared}. Continuous metric learning instead couples distances to
graded targets \c
+
+ MMCTL contributes: (i)~a joint annotation vector with normalization and task weights
(Eqs.~\eqref{eq:ann-norm}--\eqref{eq:ann-concat}); (ii)~a margin-free continuous triplet with adaptive gap
$D(\mathbf{y}_a,\mathbf{y}_q)-D(\mathbf{y}_a,\ma
```

Hunk baseline L315–317 → current L315–317 (replace)

```
– % TODO(B2): update numeric tables after Agent A replication deltas (results/replication_awir*).
– % TODO(B3): insert compute-cost, missing-modality, significance, and failure-analysis text/figures from
results/new_experiments_summary.md (no placeholder empty sections).
–
+
+ \paragraph{Reproducibility.}
+ AWIR structure scores in Table~\ref{tab:latent_alignment} are reproduced with released pretrained embeddings via
\texttt{demo\_metrics.py}; absolute mean drift is at most $0.13$ (MMCTL box), with ranking unchanged. Absolute
DeepSet downstre
```

Hunk baseline L462–461 → current L462–463 (insert)

```
+ Paired Wilcoxon tests across the $8$ AWIR runs confirm that MMCTL improves structure scores over the strongest
competing method on each annotation source (class vs.\ Concat.; box and position vs.\ RGB; all $p{=}0.0078$).
Supporting linear p
```

```
+
```

Hunk baseline L546–548 → current L548–548 (replace)

```
– An ablation study evaluates how varying the strength of semantic supervision influences the learned MMCTL
representation on NEON. The classification annotation weight is varied from 0.5 to 2.0 in increments of 0.25,
while the box and height
–
– \textbf{Results.} Fig.~\ref{fig:ablation2} shows that moderate classification weighting produces the strongest
overall performance while extreme values introduce trade-offs. Classification improves as semantic weight
increases from low valu
+ We vary the NEON classification weight from $0.5$ to $2.0$ (box and height fixed at $1.0$).
Fig.~\ref{fig:ablation2} shows moderate semantic weight yields the best overall trade-off: classification and box
improve into a mid-range setting,
```

Hunk baseline L557–556 → current L557–575 (insert)

```
+ \subsection{Missing Modality, Compute, and Failure Analysis}
+ \label{subsec:additional-analyses}
+
+ \paragraph{Missing modality.}
+ As a frozen-embedding probe on released AWIR test embeddings, we zero one $1024$-d half of the $2048$-d MMCTL
joint vector (RGB$|$thermal), L2-normalize, and retrain linear probes. Full joint proxy accuracy is
$93.44\pm1.37\%$. Dropping the
+
+ \paragraph{Triplet compute.}
+ With batch size $32$, modality concatenation yields $64$ embeddings and an upper bound of $249{,}984$ ordered
triplets per batch ($\approx 2.0$M per epoch at eight steps) before the annotation-order mask; actual positives
are data-dependent
+
+ \paragraph{Cross-modal retrieval.}
+ Splitting the joint embedding into modality halves and performing cosine nearest-neighbor retrieval gives rank-1
\emph{class} match of $80.8\%$, but rank-1 \emph{identity} match of only $7.5\%$. Failures concentrate on adjacent
livestock co
+
+ \begin{figure}[!htb]
+ \centering
+ \includegraphics[width=\columnwidth]{figures/awir_retrieval_failures.png}
+ \caption{AWIR cross-modal NN failures (RGB half $\rightarrow$ thermal half) for MMCTL joint embeddings.
Mismatches are typically adjacent livestock classes rather than arbitrary semantics; identity rank-1 is only
$7.5\%$, so we do not c
+ \label{fig:retrieval_failures}
+ \end{figure}
+
```

Hunk baseline L558–560 → current L577–577 (replace)

```
– This work frames multi-modal representation learning as aligning embeddings across sensing modalities using
annotation-defined similarity rather than task-loss geometry alone. MMCTL builds a shared latent space by coupling
modality-specific
– % TODO(B3): optionally expand conclusion with compute/missing-modality/failure takeaways once Agent A artifacts
land.
–
+ This work frames multi-modal representation learning as aligning embeddings across sensing modalities using
annotation-defined similarity rather than task-loss geometry alone. MMCTL builds a shared latent space by coupling
modality-specific
```

Diff stats: 29 added/changed lines shown in red; 12 removed baseline lines shown in green.

Full comment checklist: `results/REVIEW_COMMENTS_TRACKER.md`. Numeric results: `results/RESULTS_TRACKER.md`.